

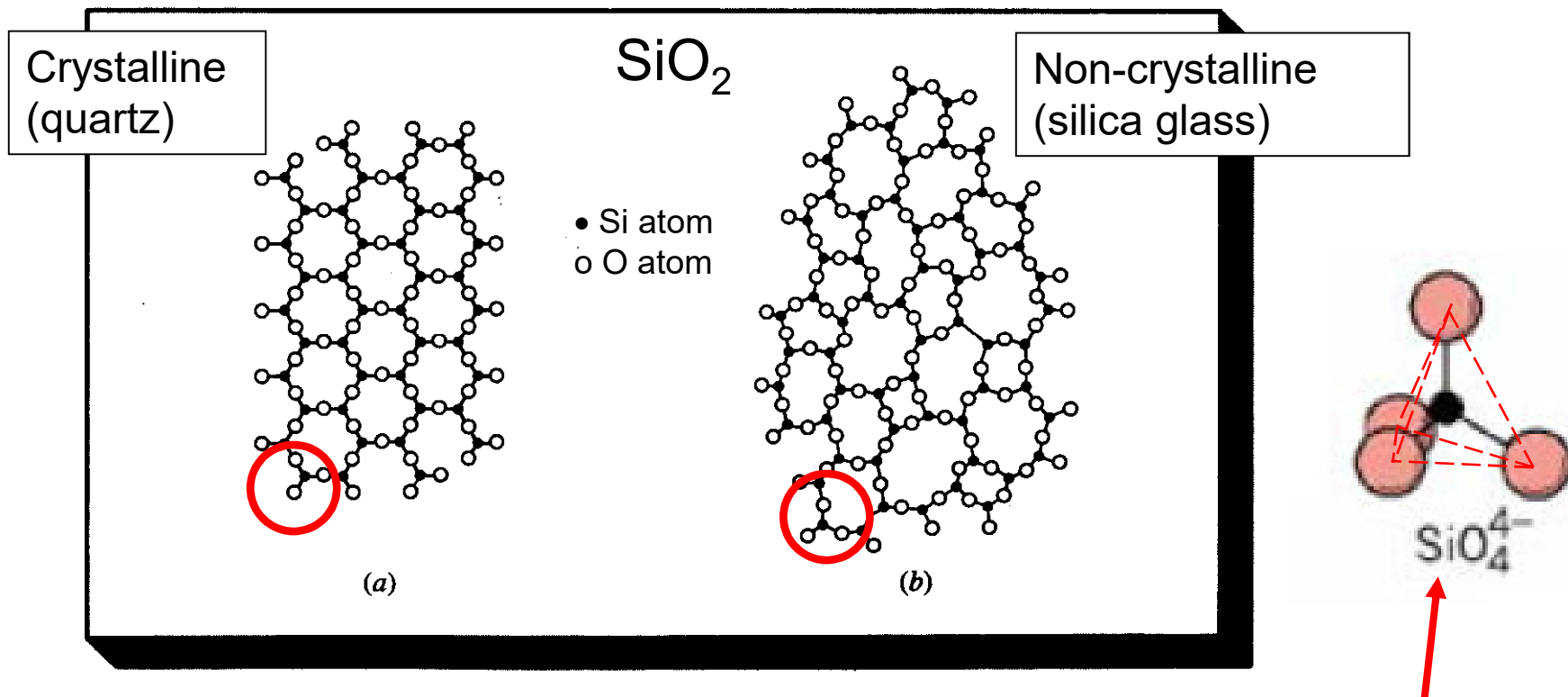
GLASS

and related products

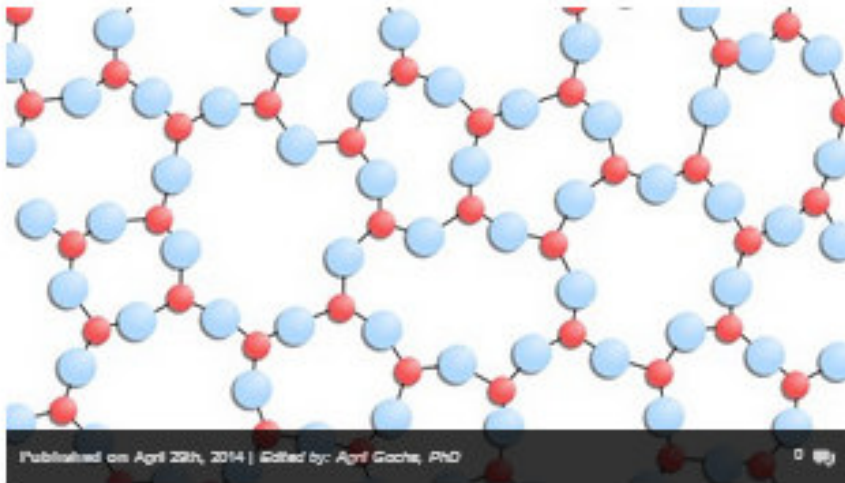
Glass structure

Glasses exhibit an **AMORPHOUS STRUCTURE**:

- they have a disordered structure (non-crystalline);
- there is no long-range atomic order;
- there is short-range atomic order.



Note that in silica glass the short-range order is only represented by the **[SiO₄]⁴⁻ tetrahedral unit** (1 atom of Si surrounded by 4 atoms of O, forming a “pyramid”)



New energy landscapes reveal glass states have rough fractal basins

Published on April 29th, 2014 | Edited by: April Gocha, PhD

<http://ceramics.org/ceramic-tech-today/new-energy-landscapes-reveal-glass-states-have-rough-fractal-basins>

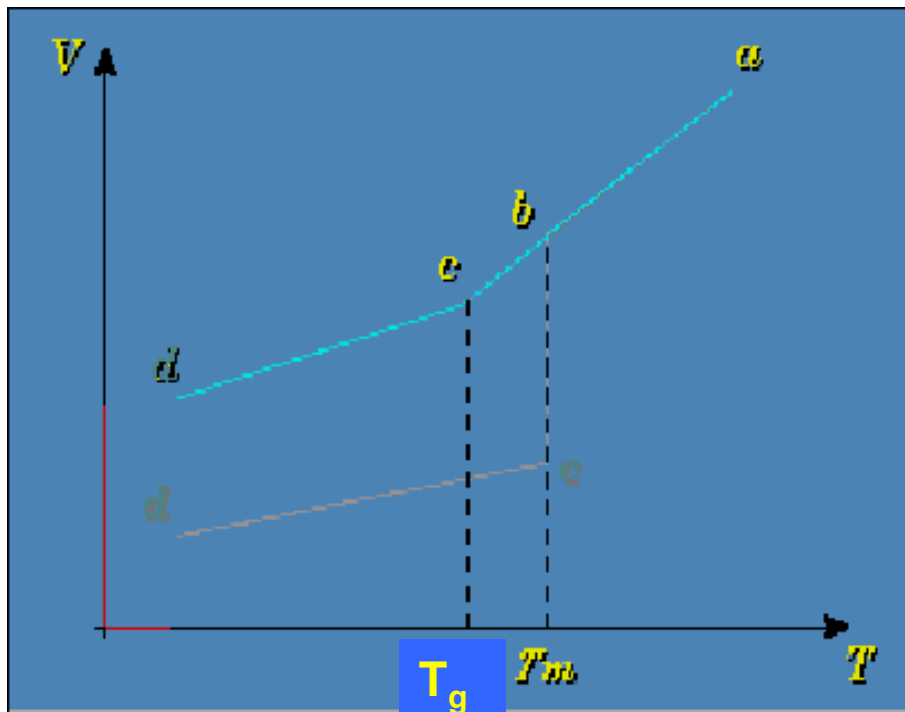
Glass is a mystery.

That's because glasses are amorphous and lack the degree of order achieved in crystalline structures. In a glass, molecules are held in place by their neighbor molecules in less predictable and disordered configurations.

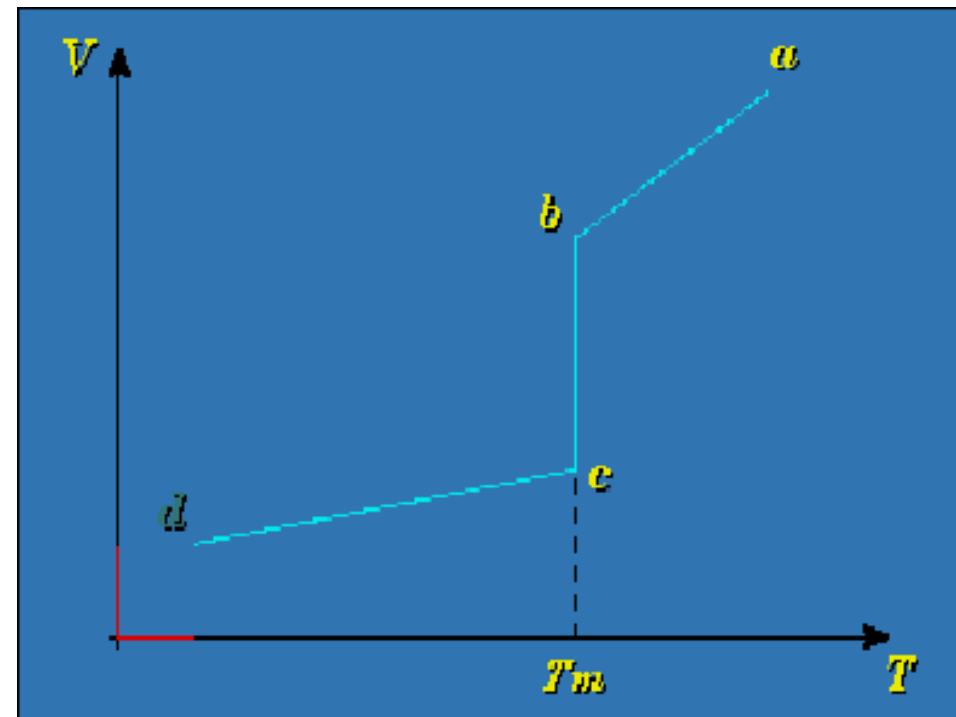
Because of these interesting features, the thermodynamics of glass states are not described by a simple model. But research from Duke University offers insights that may be help to unlock these mysteries by providing a new [energy landscape](#) of glasses. The work shows that the landscape, which maps all possible energy positions of the glass molecules, is much rougher than previously believed.

According to a Duke [press release](#), lead author [Patrick Charbonneau](#) says, "There have been beautiful mathematical models, but with sometimes tenuous connection to real, structural glasses. Now we have a model that's much closer to real glasses." Charbonneau, a professor of chemistry and physics at Duke, and a team of scientists used mathematical models and theory to analyze how glass molecules behave,

Glass solidification



Crystal solidification



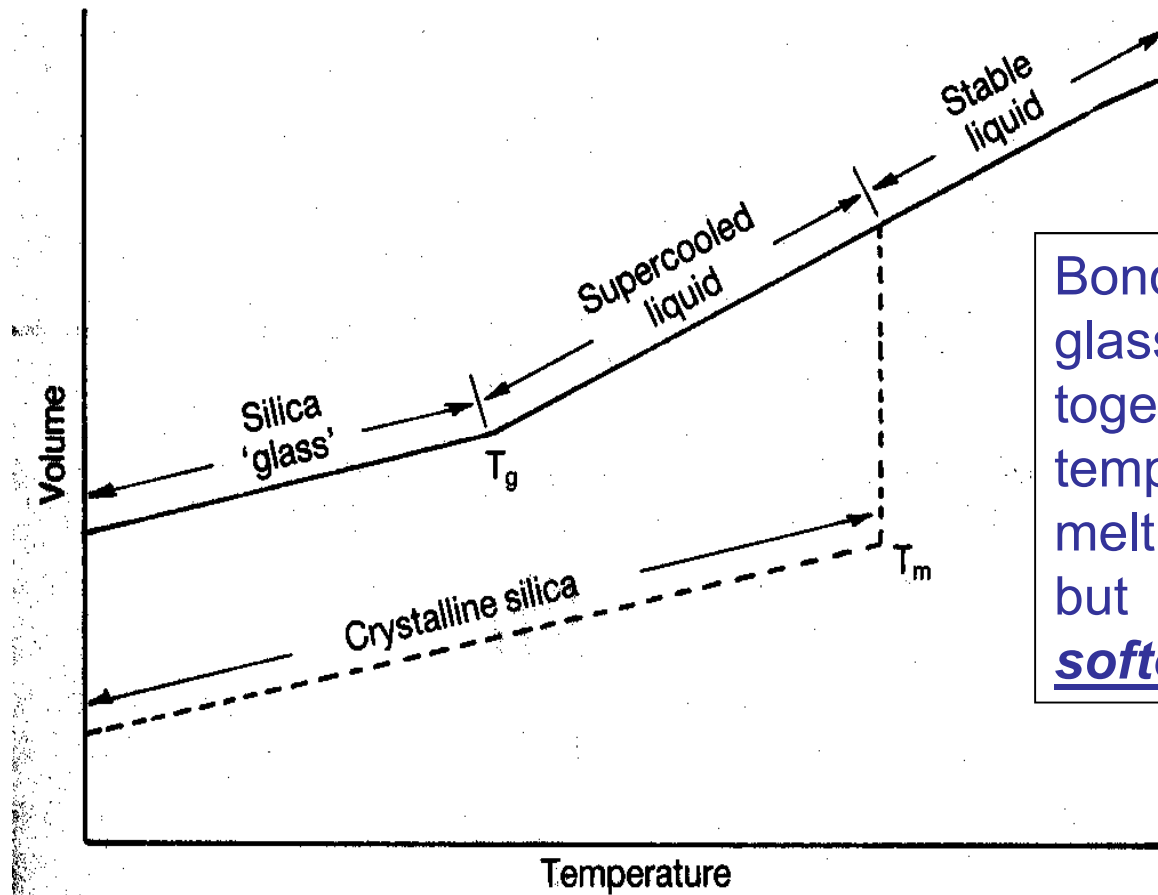
Glasses do not solidify in the same sense of crystalline materials: upon cooling, glasses become more and more viscous and there is no definite temperature at which the liquid transforms into solid as for crystalline materials.

GLASSES:

glass transition temperature T_g
above which glass
progressively softens

CRYSTALLINE SOLIDS:

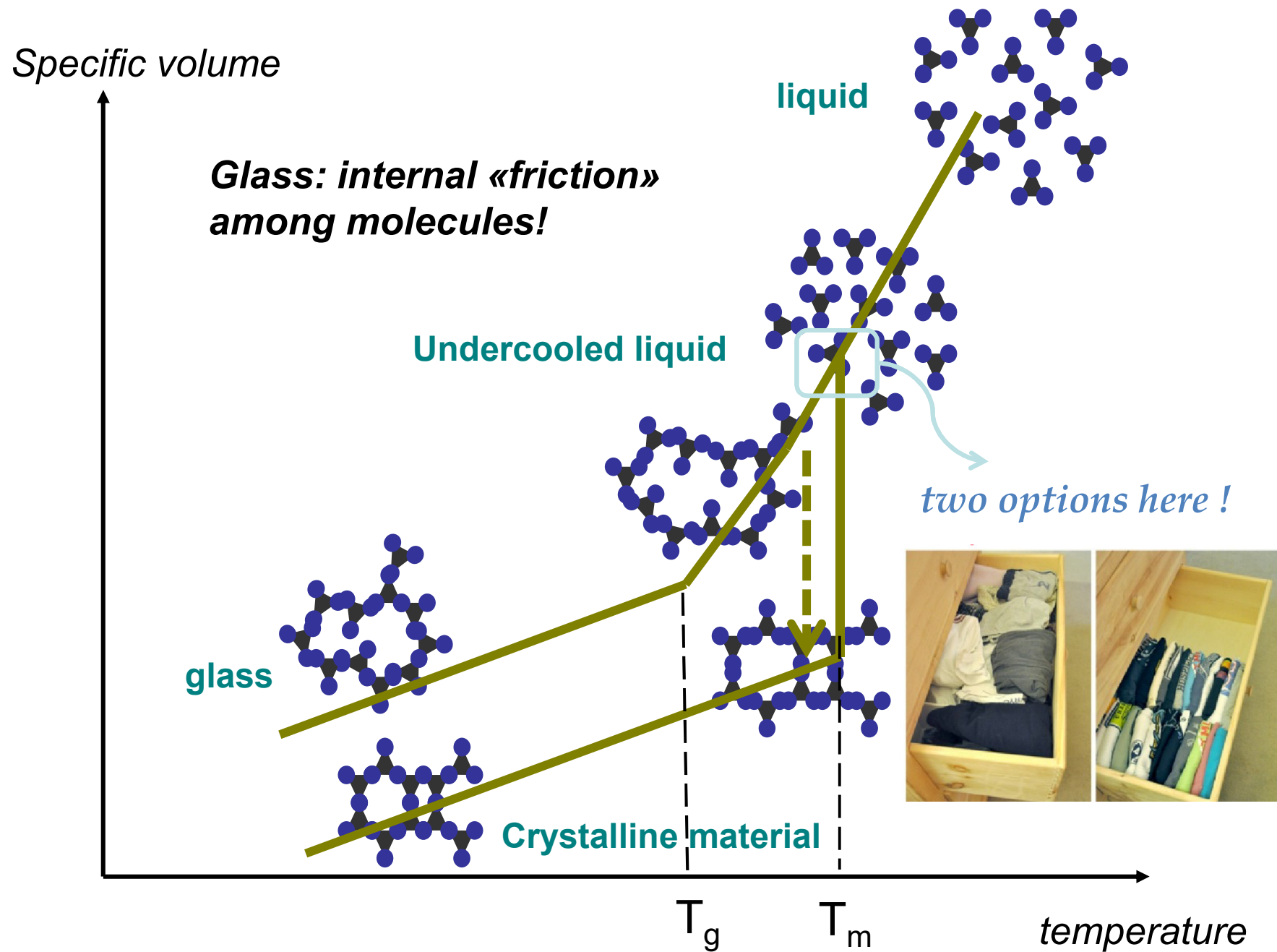
melting temperature T_m



Bonds in amorphous solids (e.g. glasses) do not break all together at a certain temperature T_m : an exact melting point cannot be defined, but the amorphous solids *soften step by step*.

T_g can be practically viewed as the temperature at which glass starts to change from a rigid state to a “rubber-like state”.

In glasses, we can define T_m as “liquid temperature”, i.e. the temperature above which the material becomes a stable liquid (it completely changes from a “rubber-like” state to a liquid state).



Glass transition temperature

Most commercial glasses are obtained by rapid cooling of a viscous liquid below a typical “freezing” temperature, known as **glass transition temperature, T_g** .

Such T is fundamental to define the properties of a certain glass:

- It is in general around 2/3 of the melting $T \rightarrow T_g = 2/3 T_m$
- above T_g the glass has a rubber-like behaviour
- below T_g the glass is a rigid and brittle material

Types of amorphous materials

General definition of "glass": any substance that, under suitable conditions, can turn into the amorphous state

1. *Ceramics:* - inorganic oxides → traditional glasses (SiO_2 , TeO_2 , etc.)
- inorganic fluorides → fluoride glasses
- inorganic chalcogenides → chalcogenide glasses
2. *Metals:* amorphous metals (obtained by casting molten metal on thin tapes with cooling speed of 10^7 °C/s)
3. *Organic compounds:* glycerin, glucose (caramel)
4. *Polymers:* in polymers amorphous zones are always present (PET), some polymers are almost completely amorphous (e.g. PMMA)
5. *Some native elements like S, Se, P*

Glass formation

Glass-forming liquids: upon cooling, viscous molten material (mixtures of inorganic oxides such as SiO_2 , CaO , Na_2O , MgO ...) reaches a temperature below the solidification temperature without crystallization → there is no time for crystalline structure formation during conventional cooling of a glass-forming liquid.

This occurs as atomic arrangement in liquid material is blocked by high viscosity: the random atomic structure of the liquid is “frozen” in the solid state out of thermodynamic equilibrium.

Example: viscosity of SiO_2 fused at 1720°C is about 10^8 times higher than that of H_2O at 20°C !!!

Amorphous structures are thermodynamically metastable (they do not represent the configuration at the lowest energy); therefore, they progressively will move into crystalline structures according to thermodynamic and kinetic conditions (DEVITRIFICATION can take several thousands of years at room temperature, but this process can be accelerated by thermal treatment below T_m).

Glasses vs. ceramics?

Note that **glasses** formally belong to the class of CERAMIC MATERIALS.

In fact, we can divide ceramics into 3 sub-families:

- **FULLY CRYSTALLINE CERAMICS** (e.g. most of **rocks and stone materials** used in construction, that can be considered as “natural ceramics”; **cements**, that can be considered “artificial ceramics”; advanced or technical ceramics such as **alumina** and **zirconia**);
- **GLASSES** (completely amorphous, non-crystalline materials);
- **GLASS-CERAMICS**, in which one or more crystalline phases are embedded in an amorphous matrix of residual glass; **devitrification** can be deliberately carried out on starting glasses by thermal treatment to impart better properties to the material (for instance, glass-ceramics exhibit superior mechanical properties, such as strength, hardness and stiffness, in comparison to their parent glasses).

Glass constituents (inorganic oxides)

NETWORK FORMERS:

(e.g. SiO_2 , B_2O_3):
oxides able to form
glassy structure

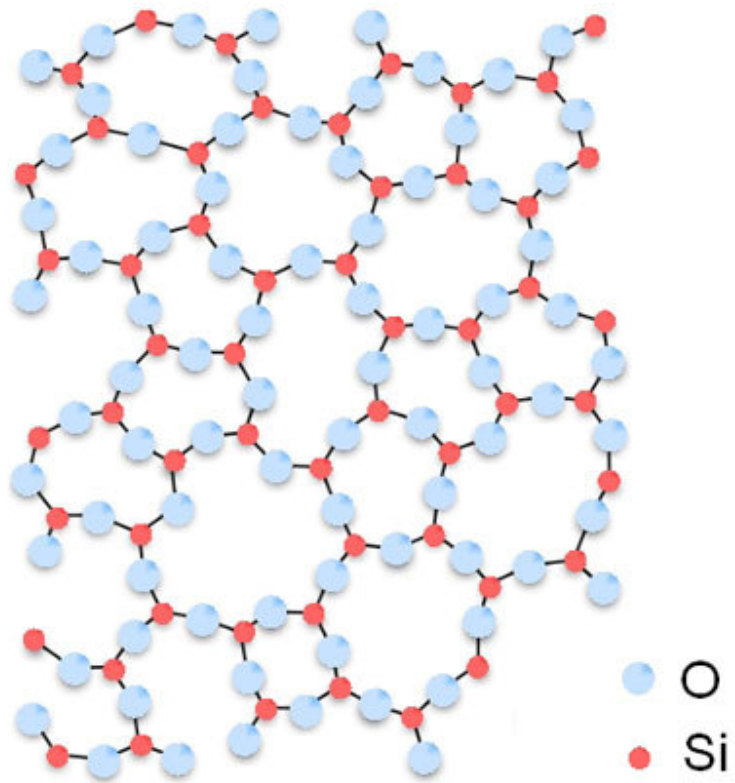
INTERMEDIATES:

(e.g. TiO_2 , Al_2O_3): oxides
that do not form the network
but can substitute atoms of
the forming network, thereby
becoming part of the network
and stabilizing it

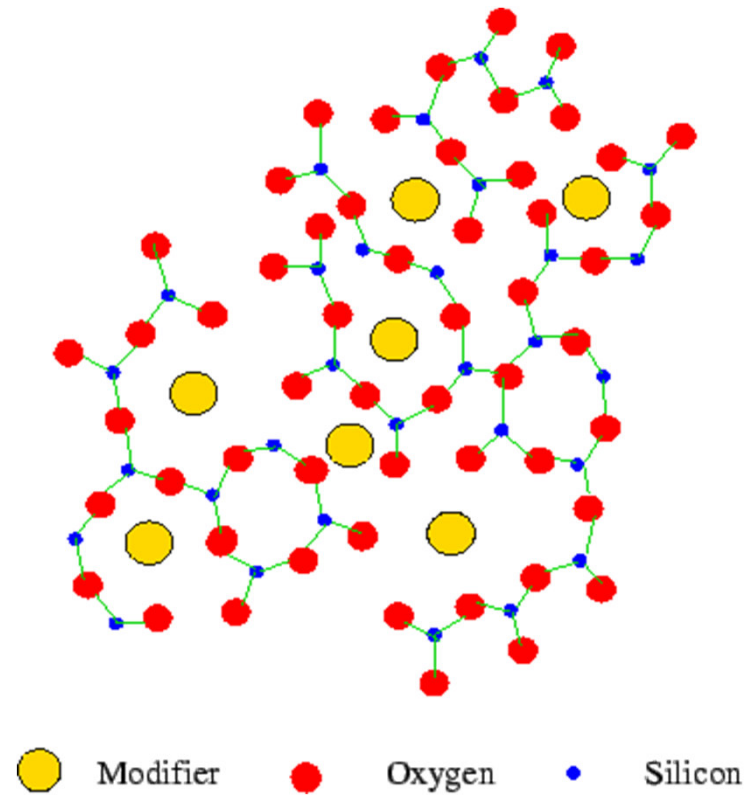
NETWORK MODIFIERS:

(e.g. CaO , Na_2O):
oxides that are not able
to form glassy
(polyhedral) networks
but their cations are
incorporated into the
glass network (e.g. in
the SiO_2 -based one)

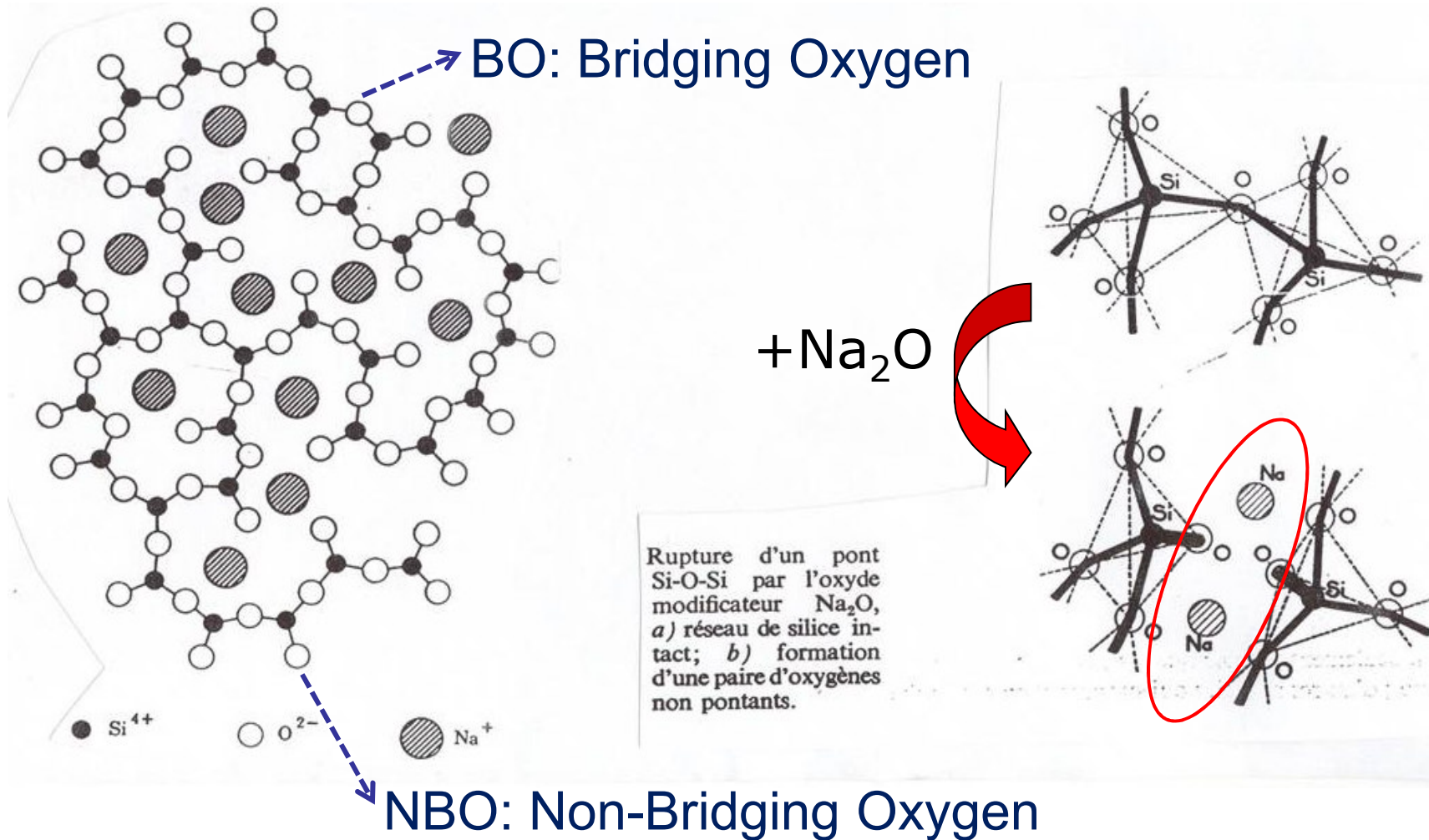
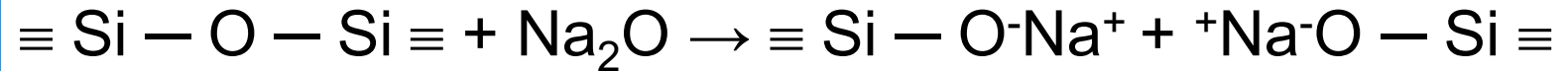
NETWORK FORMERS

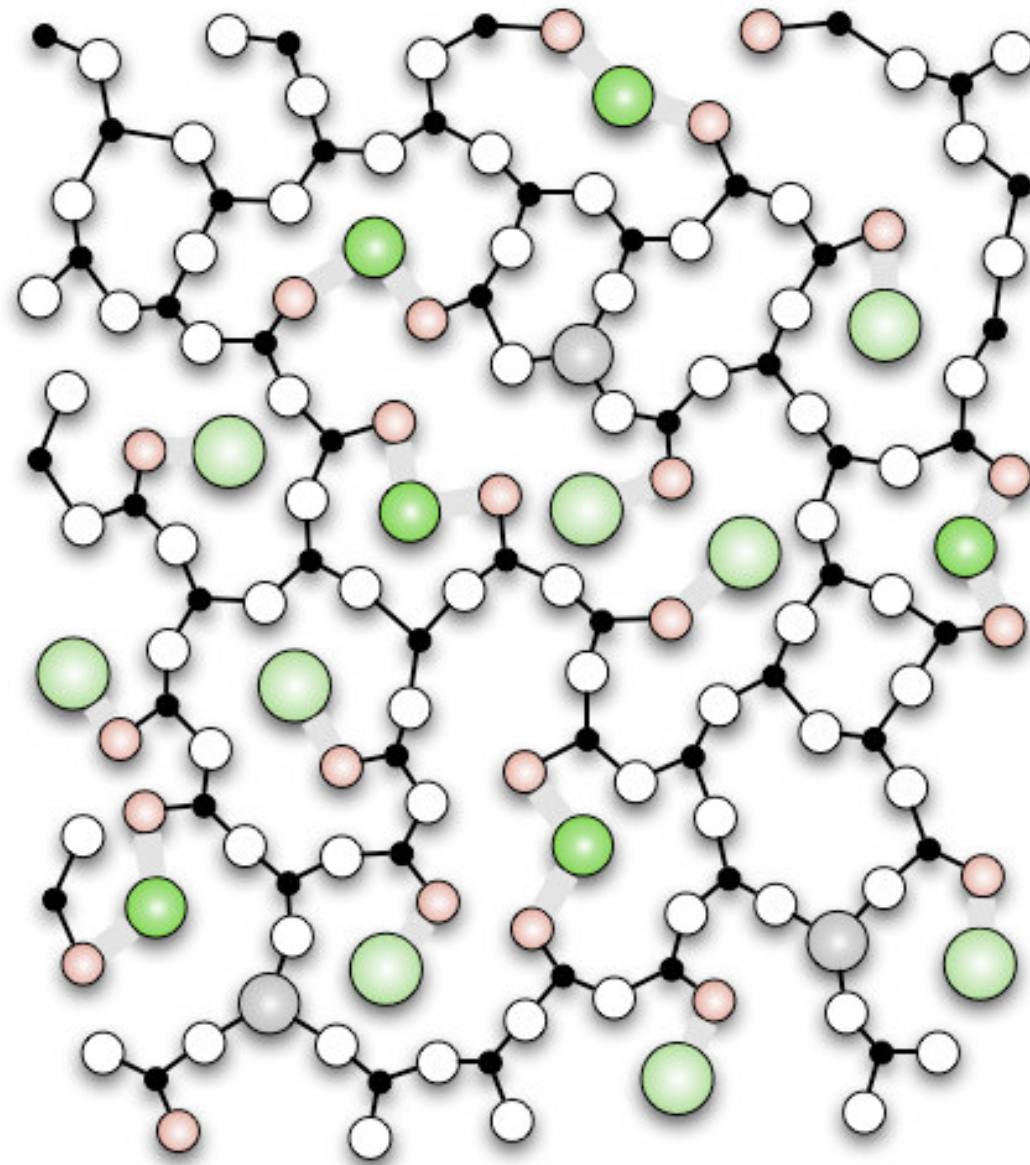


NETWORK MODIFIERS



Glass structure: effect of alkali oxide





Network of a
quaternary (4-oxide) glass
 belonging to the
 $\text{SiO}_2\text{-Al}_2\text{O}_3\text{-NaO-CaO}$
 system

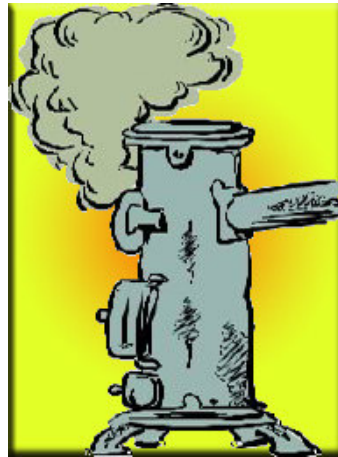


Glass synthesis

The most commonly used method used for making glasses is the **melting-quenching** route



Reagents mixing in a crucible (e.g. SiO_2 , carbonates of Na and Ca, Al_2O_3 ...)



Melting in high-temperature furnaces (typically above 1000°C)



Pouring in moulds (e.g. for windows) or in water for very rapid cooling

Furthermore, upon cooling, soft glass can undergo **pressing** and/or **air blowing** (for instance to produce bottles) or be **drawn in fibres** (for instance to be used as reinforcing phase in polymer-matrix composites).

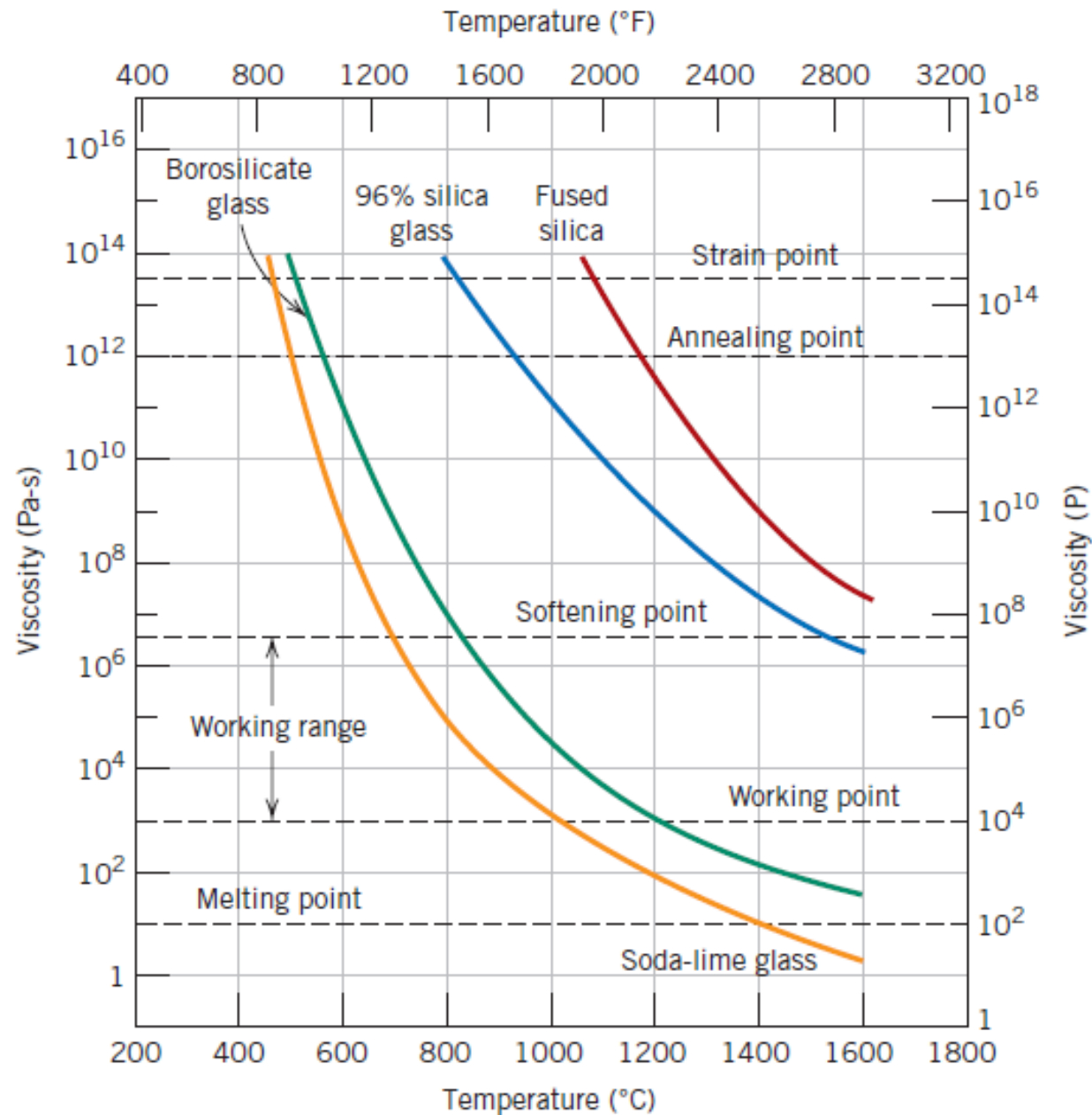


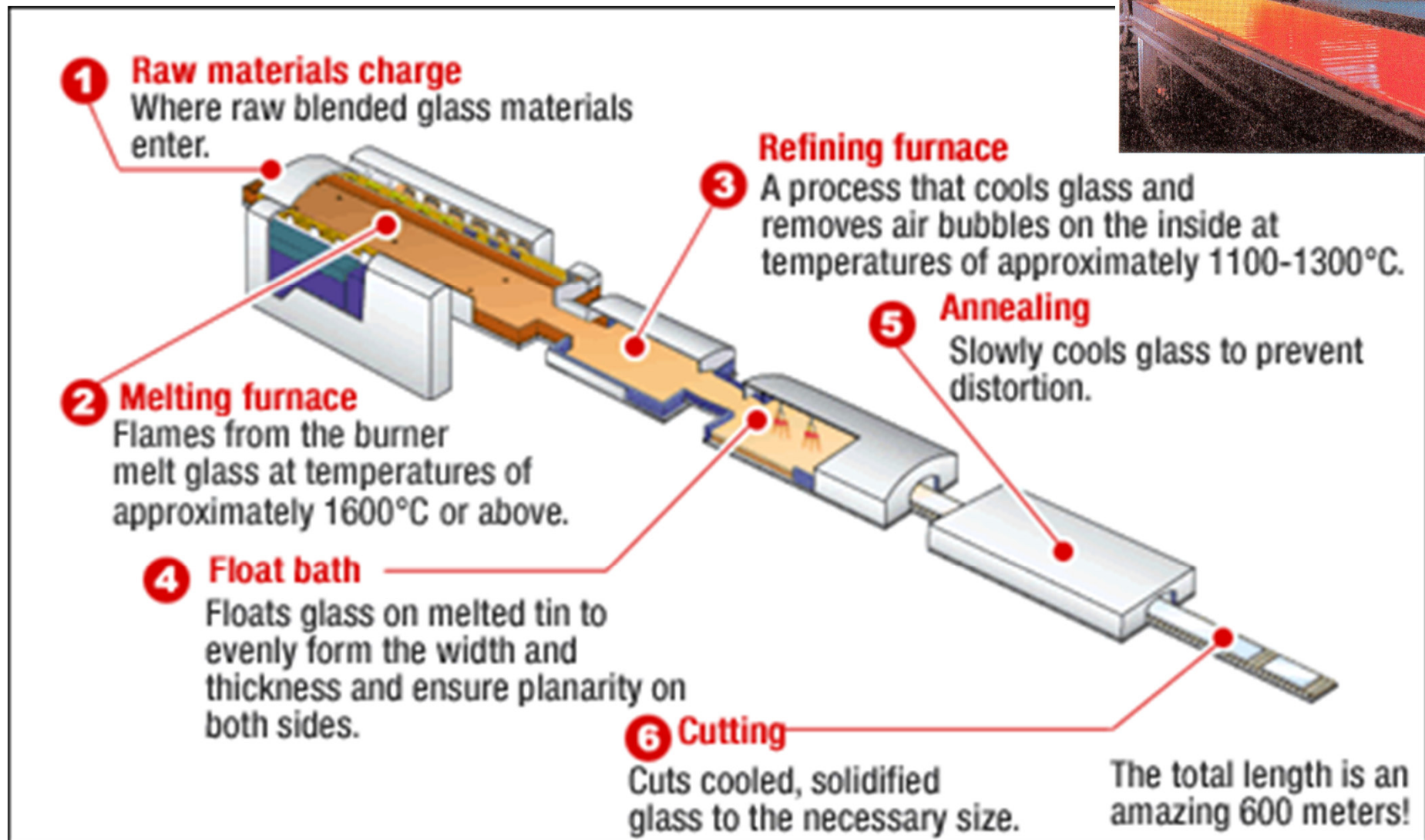
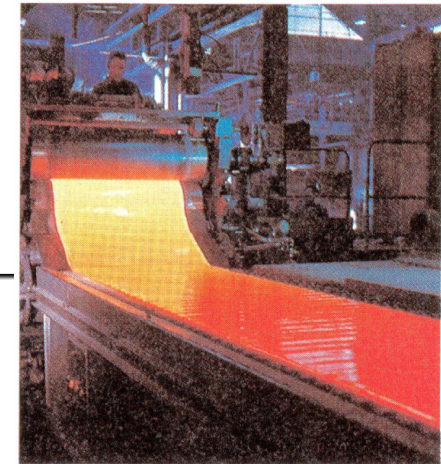
Figure 13.7

Logarithm of viscosity versus temperature for fused silica and three silica glasses. (From E. B. Shand, *Engineering Glass, Modern Materials*, Vol. 6, Academic Press, New York, 1968, p. 262.)

Conventional values of viscosity

$$\text{Strain point} < T_g < \text{Annealing point}$$

Float glass production → use of a **melted TIN** bath



Coloured glasses

We can add chromophore ions in form of small quantities of special **metal oxides** to the basic glass composition

Ioni cromofori.

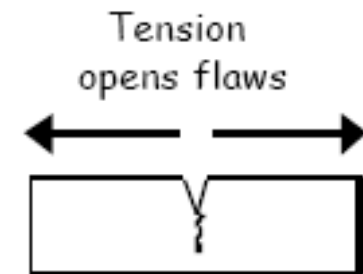
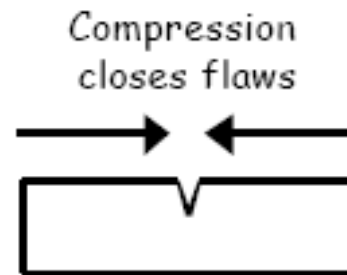
Ioni cromofori	Coordinazione	Colorazione
Cu^{2+}	$[\text{Cu}^{\text{II}}\text{O}_6]$	blu pallido
Cu^+		incolore
Ti^{3+}	$[\text{Ti}^{\text{III}}\text{O}_6]$	violetto-marrone
V^{3+}	$[\text{V}^{\text{III}}\text{O}_6]$	verde
V^{5+}	$[\text{V}^{\text{V}}\text{O}_4]$	incolore
Cr^{3+}	$[\text{Cr}^{\text{III}}\text{O}_6]$	verde
Cr^{6+}	$[\text{Cr}^{\text{VI}}\text{O}_4]$	giallo
Mn^{2+}	$[\text{Mn}^{\text{II}}\text{O}_6]$	incolore
Mn^{3+}	$[\text{Mn}^{\text{III}}\text{O}_6]$	violetto
Fe^{3+}	$[\text{Fe}^{\text{III}}\text{O}_4]$, $[\text{Fe}^{\text{III}}\text{O}_6]$	giallo
Fe^{2+}	$[\text{Fe}^{\text{II}}\text{O}_6]$	blu
$\text{Fe}^{3+} + \text{Fe}^{2+}$		verde
Co^{2+}	$[\text{Co}^{\text{II}}\text{O}_4]$	blu intenso
Co^{2+}	$[\text{Co}^{\text{II}}\text{O}_6]$	rosa (in vetri borati)
Co^{3+}	$[\text{Co}^{\text{III}}\text{O}_4]$	verde
Ni^{2+}	$[\text{Ni}^{\text{II}}\text{O}_6]$	giallo (in vetri contenenti Li_2O)
Ni^{2+}	$[\text{Ni}^{\text{II}}\text{O}_4]$	blu (in vetri contenenti K_2O)
Ni^{2+}	$[\text{Ni}^{\text{II}}\text{O}_6] + [\text{Ni}^{\text{II}}\text{O}_4]$	bruno-grigio

Mechanical behaviour of glasses

Usually glasses are not used as structural materials, but they can sustain loads (windscreen and glass windows are exposed to wind, impacts...)

Glasses are brittle materials!

Stress concentration near pores or macroscopic defects (cracks, flaws)
→ brittle materials show better strength under compressive load



Mechanical properties of glasses, together with other physical properties (e.g. optical properties, transparency...) **STRONGLY** depend on **GLASS COMPOSITION** (= the type and amount of the **oxides** we use for making the glasses)

Typical features of glasses

Density: 2.2-6.3 g/cm³; for common glasses 2.5 g/cm³

Hardness: 5.5-6.5 Mohs (for instance higher than metals, but lower than quartz and granite)

Elastic (Young's) modulus: $E = 75 \text{ GPa}$

Compressive strength: $\sigma = 1000 \text{ MPa}$

Tensile/flexural strength: $\sigma = 40 \text{ MPa}$ for common glasses; up to 200 MPa for special glasses

Thermal expansion coefficient: $9 \times 10^{-6} \text{ K}^{-1}$ (lower than metals but quite high: for instance, a 2-m long glass fiber shows elongation of 0.72 mm because of heating of 40 °C)

Thermal conductivity: very low

Thermal shock resistance: low for common glass, good for silica glass

Refractive index: $n = 1.5$ for common glasses

The most common glasses are **silica-soda-lime** (SiO_2 - Na_2O - CaO system) glasses:

- Low cost
- Easy processing and forming
- Good resistance to devitrification and stability in water
- Al_2O_3 addition increases glass durability
- Uses: house windows, car/train windows, windscreens, light bulb...
- Poor resistance to heat and to sudden changes of temperature

Table 9.1 Room-Temperature Yield Strength and Plane Strain Fracture Toughness Data for Selected Engineering Materials

Material	Yield Strength		K_{Ic}	
	MPa	ksi	MPa $\sqrt{m}$	ksi $\sqrt{in.}$
Metals				
Aluminum Alloy ^a (7075-T651)	495	72	24	22
Aluminum Alloy ^a (2024-T3)	345	50	44	40
Titanium Alloy ^a (Ti-6Al-4V)	910	132	55	50
Alloy Steel ^a (4340 tempered @ 260°C)	1640	238	50.0	45.8
Alloy Steel ^a (4340 tempered @ 425°C)	1420	206	87.4	80.0
Ceramics				
Concrete	—	—	0.2–1.4	0.18–1.27
Soda-Lime Glass	—	—	0.7–0.8	0.64–0.73
Aluminum Oxide	—	—	2.7–5.0	2.5–4.6
Polymers				
Polystyrene (PS)	—	—	0.7–1.1	0.64–1.0
Polymethyl Methacrylate (PMMA)	53.8–73.1	7.8–10.6	0.7–1.6	0.64–1.5
Polycarbonate (PC)	62.1	9.0	2.2	2.0

^a **Source:** Reprinted with permission, *Advanced Materials and Processes*, ASM International, © 1990.

K_{Ic} critical parameters
(defect length and stress)
above which there is failure (*all materials*)

$$K_{Ic} = Y \sigma_c (\pi a)^{1/2}$$

(ASTM E 399)

Safety glasses

Designers must choose glasses taking into account some severe rules.

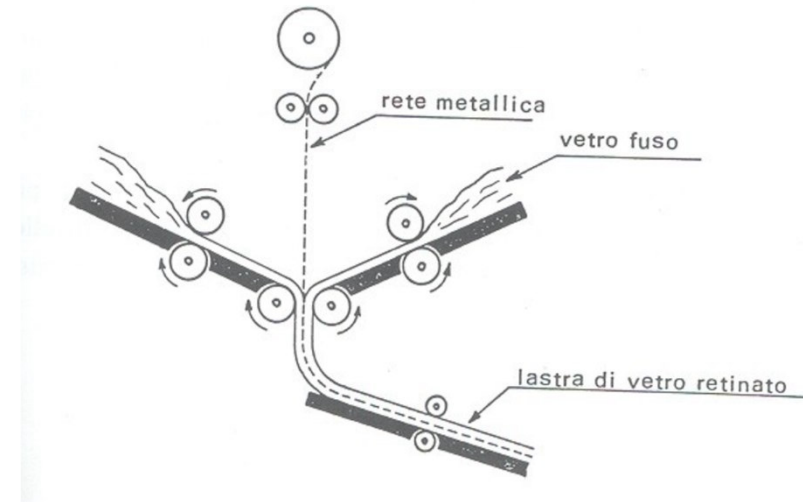
Glass sheets can be subjected to:

- Dynamic loads (wind, pressure of the crowd)
- Static loads (own weight, snow, hydrostatic pressure)
- Accidental loads (hail, vibrations, torsion, seismic phenomena)
- Impact with soft body (people and animals, usually accidental and with low intensity)
- Impact with hard body (stones, burglary tools)
- Impact with projectile (fire-arms)
- Thermal shocks (fire)

ARMoured GLASS

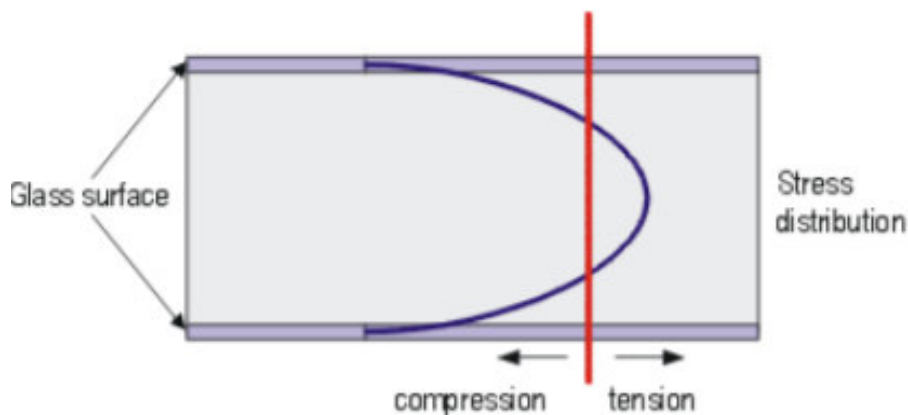
Glass containing inside a metallic network.

Mechanical strength is not particularly enhanced (slight increase of bending resistance) but there is **flame propagation delay in case of fire** → if glass softens or breaks, metallic network allows to maintain glass in its position for certain period.



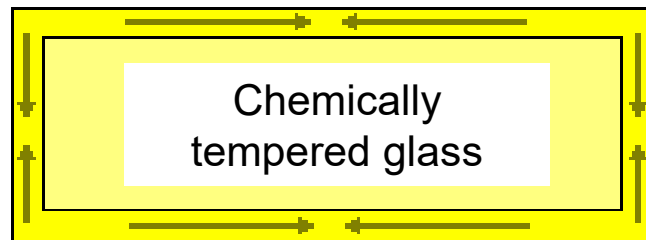
TEMPERED GLASS

Using physical (thermal → fast solidification of the surface) or chemical processes, **permanent compressive stress on the surface** and **tensile stress in the core** are induced → superior resistance to mechanical impacts in windows

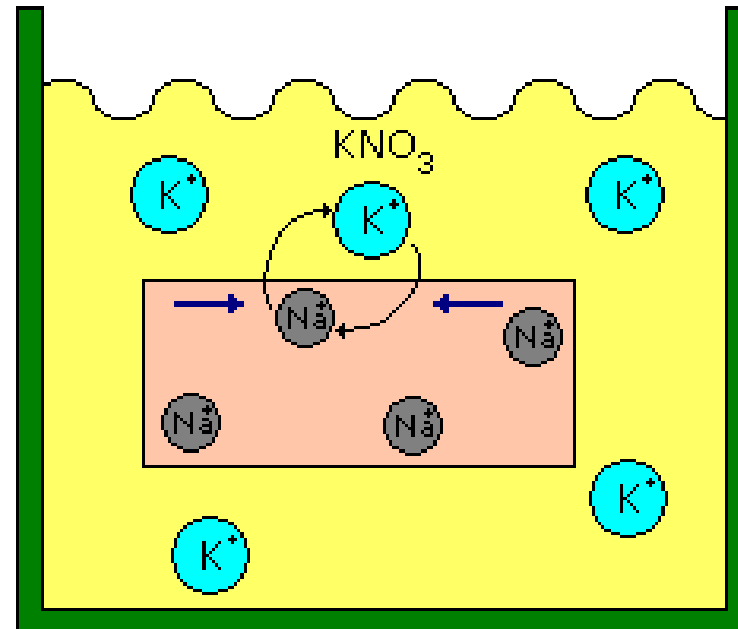




http://www.corning.com/gorillaglass/chemical_strengthening.aspx



K^+ ions are larger than Na^+ ones and induce compressive state of stress on the glass ion exchanged surface.





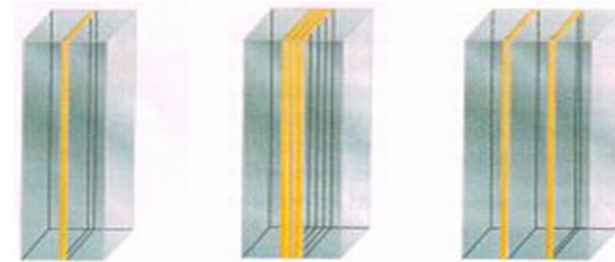
Float glass



Tempered glass

- Good shock resistance
- Rupture is localized
- Glass fragments remain attached to the polymeric sheet

MULTILAYERED GLASS



Multilayered glass

Foamed glass (or cellular glass)

Used for thermal and acoustic insulation, it has a **cellular, water-proof and gas-proof structure**.

Main features: good insulation ability, low density, high chemical stability, satisfactory mechanical performances (adequate compressive strength).

It is characterized by a high amount of small, closed (**non-interconnected**) PORES → suitable for **thermal and acoustic insulation** (air is a bad heat and sound conductor).

It is used for civil building (load-bearing substructures for roof, paving, walls) and industrial use (high- or low-temperature plants, gas tanks, chimneys, tunnels).

Commercial name:
FOAMGLAS®

Major constituents:

SiO_2 65.77 %wt.

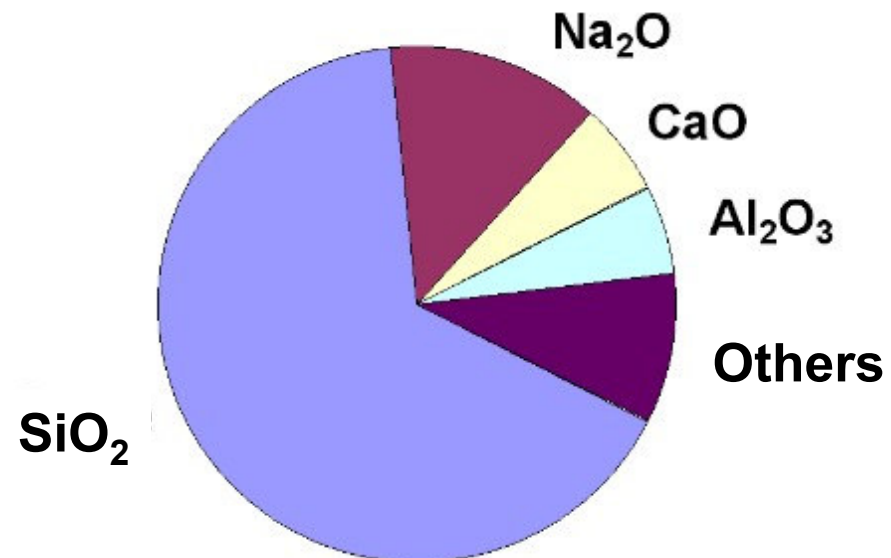
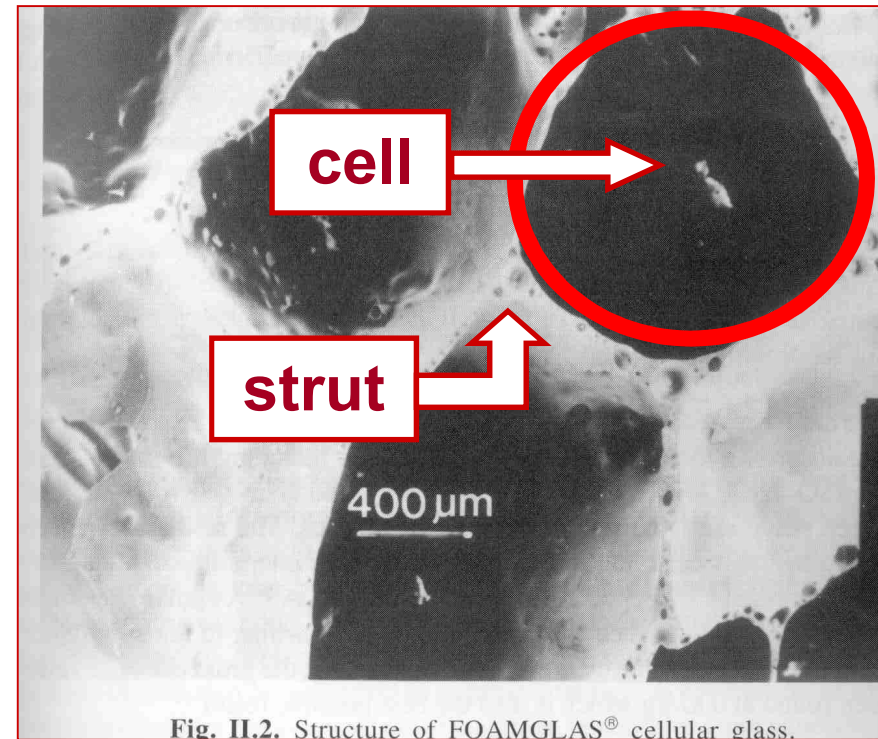
Na_2O 13.44 %wt.

CaO 5.83 %wt.

Al_2O_3 5.58 %wt.

Minor constituents:

Fe_2O_3 , MnO , K_2O , SO_3 ,
 BaO , TiO_2 , SrO , carbon
< 9.38 %wt.

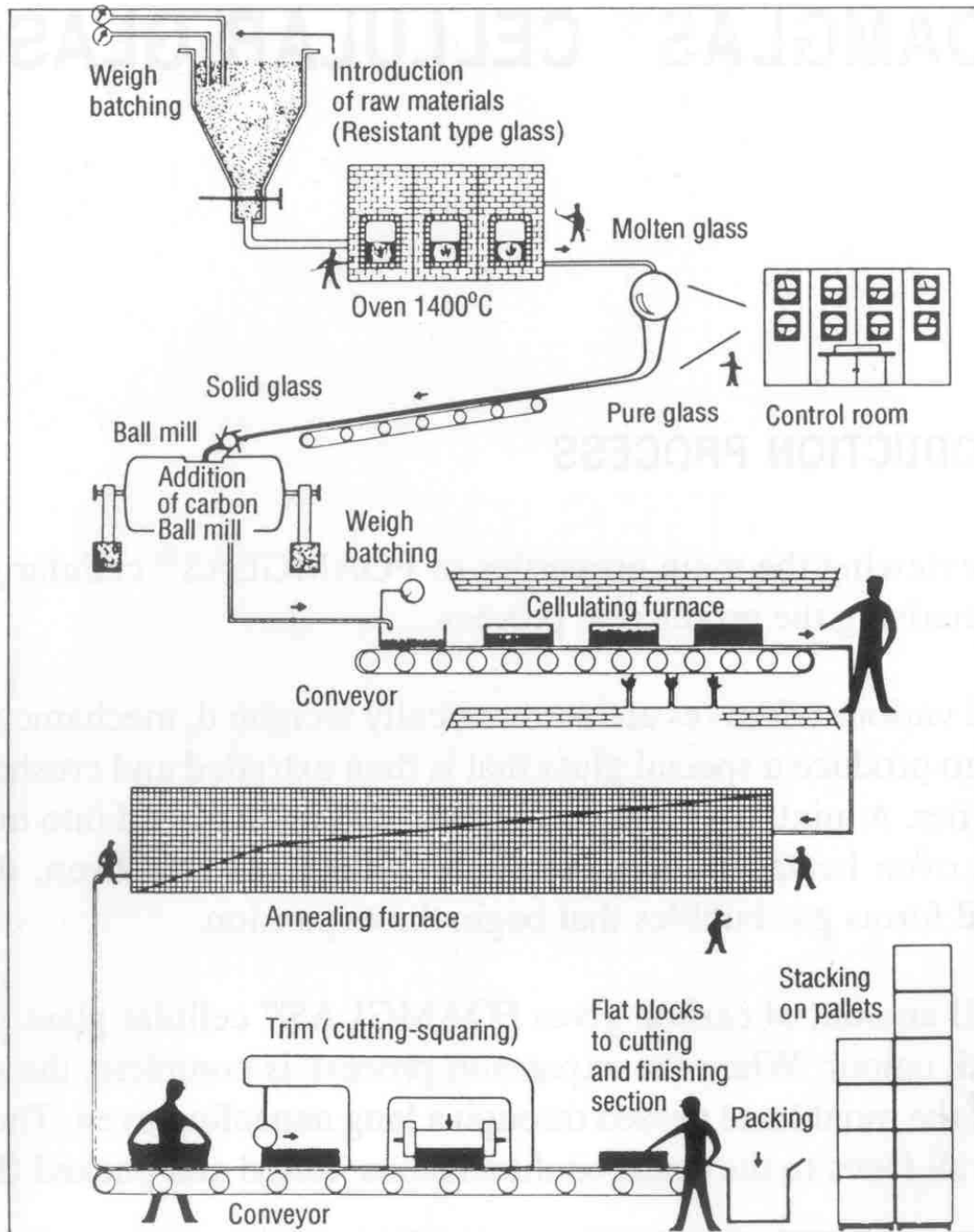


Attractive properties of FOAMGLAS®: cheap, ease of processing, eco-friendly

FOAMGLAS® is an excellent material for civil construction and insulation purposes. It is a lightweight, extremely fine-pored expanded glass with millions of **hermetically sealed pores**. Since no diffusion can take place, the material is watertight and achieves an efficient barrier against soil humidity, vapour and aggressive agents from outside.

Besides the outstanding mechanical and thermal properties of the product, foamed glass manufacture is an exemplary process for **WASTE RECYCLING** on an industrial basis. Foam glass can be manufactured fully out of waste glass, with only a minimum of virgin additives.

Foamed is especially suitable for thin-walled thermal insulations, such as for **window frames, cement bricks and insulating plasters** (FOAMGLAS® is often joined to another “traditional” material).



Foamed glass is prepared by **gas blowing** in a glass melt upon cooling, or by introducing in the starting glass composition **particular components** that, upon heating (reagents melting) **can decompose developing gas**.

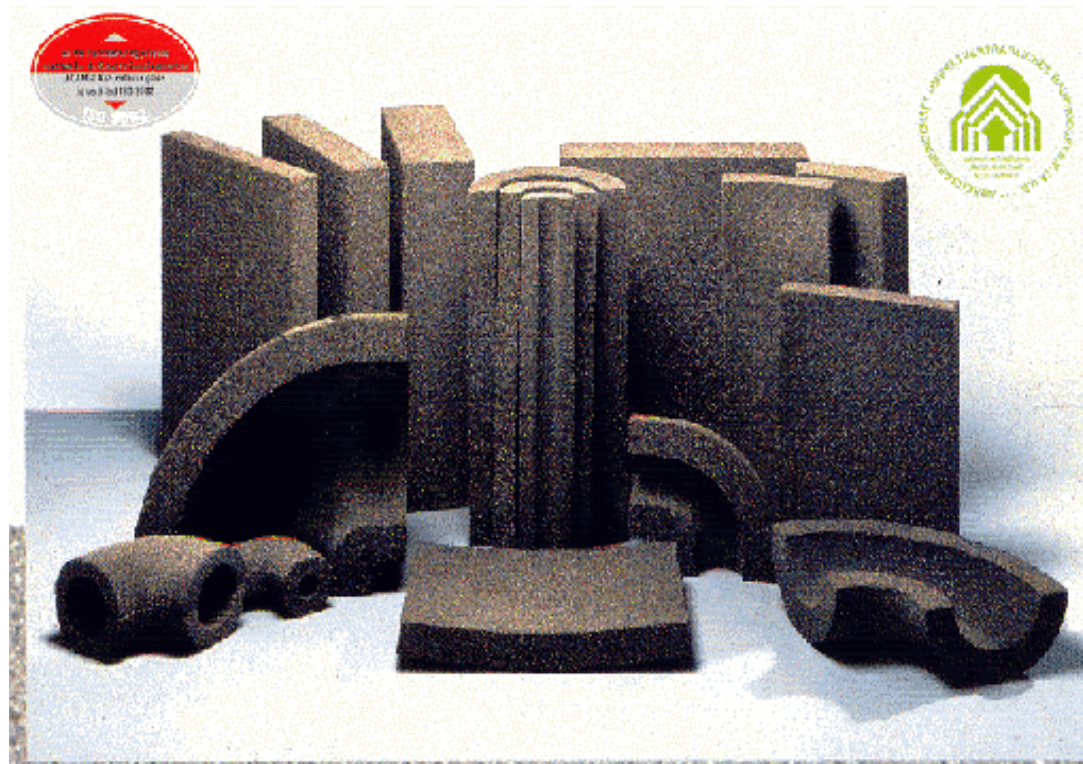
Foamed glass processing is **eco-friendly**: the majority of the starting raw material is constituted by glass from broken windscreen, recycled glass and silica sandstone.

Starting glass scraps are molten, and the produced glass is ground and milled with coal powder.

This glass/coal blend is then introduced in appropriate moulds that are only partially filled; the moulds are sealed and thermally-treated in a temperature range within 600-800 °C.

Upon heating, glass powders sinter and gas bubbles develop (coal burning-out), thus the mass occupy the whole mould volume.

- FOAMGLAS® is mainly fabricated in form of plates or panels (= assembled plates), but can be also modelled in complex shapes.
- The plates and panels are normally put in place with the aid of molten bitumen (= blend of petroleum-derived natural hydrocarbons) or by means of synthetic adhesives.
- FOAMGLAS® is manually cut and requires no special tools for installation or for the protection of the operator.



Properties and Uses of Some Glasses and Glass-Ceramics

Composition	Property	Use
<i>Glasses</i>		
Al_2O_3 , MgO , CaO , SiO_2	Translucent, chemically resistant	Window glass, bottles
PbO , SiO_2	High refractive index	Lead crystal
B_2O_3 , SiO_2 , Na_2O	Acid resistant, low expansion on heating	Pyrex
<i>Glass Ceramics</i>		
MgO , Al_2O_3 , SiO_2	Insulator with high mechanical strength at high temperatures	Spark plug insulators
CaSiO_3 , $\text{CaMgSi}_2\text{O}_6$, $\text{CaAl}_2\text{Si}_2\text{O}_8$	Wear resistant	Building materials
$\text{Li}_2\text{Si}_2\text{O}_5$	Resistant to thermal shock	Nose cones on rockets, cookware

Properties are dictated by:

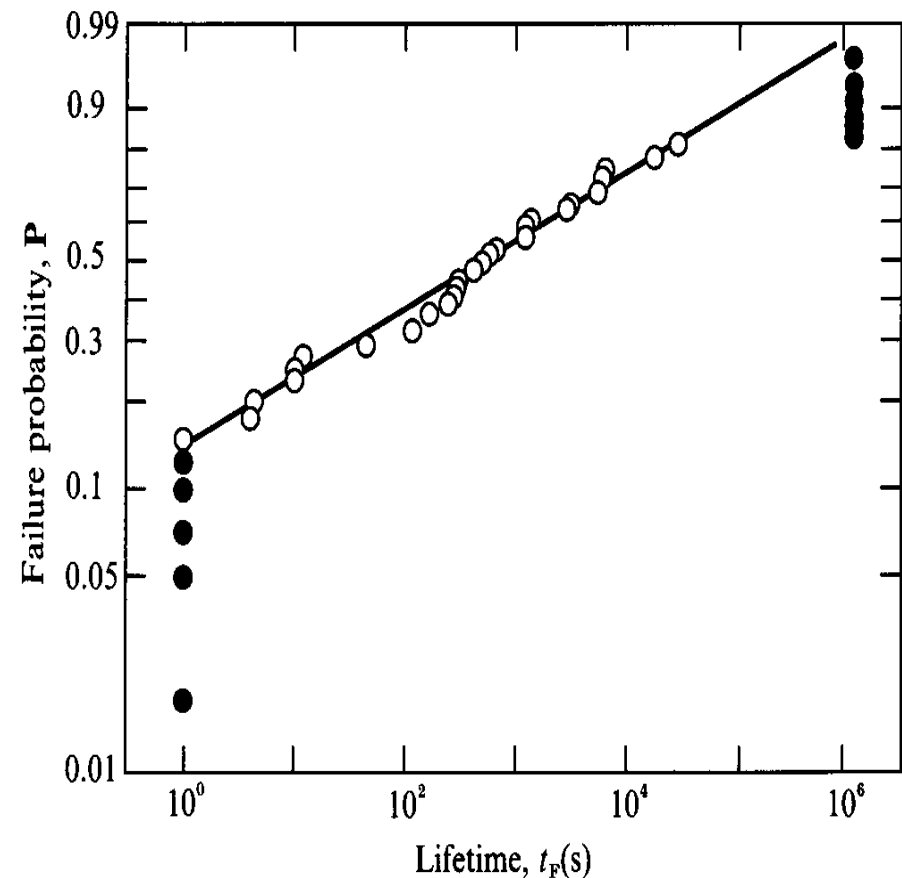
- composition (e.g. for E: $\uparrow \text{SiO}_2$, $\text{Al}_2\text{O}_3 \uparrow$ E; for CTE: $\uparrow$ modifiers $\uparrow$ CTE)
- post-processing treatments (e.g. tempering, thermal devetrification)

Mechanical resistance of glass over time

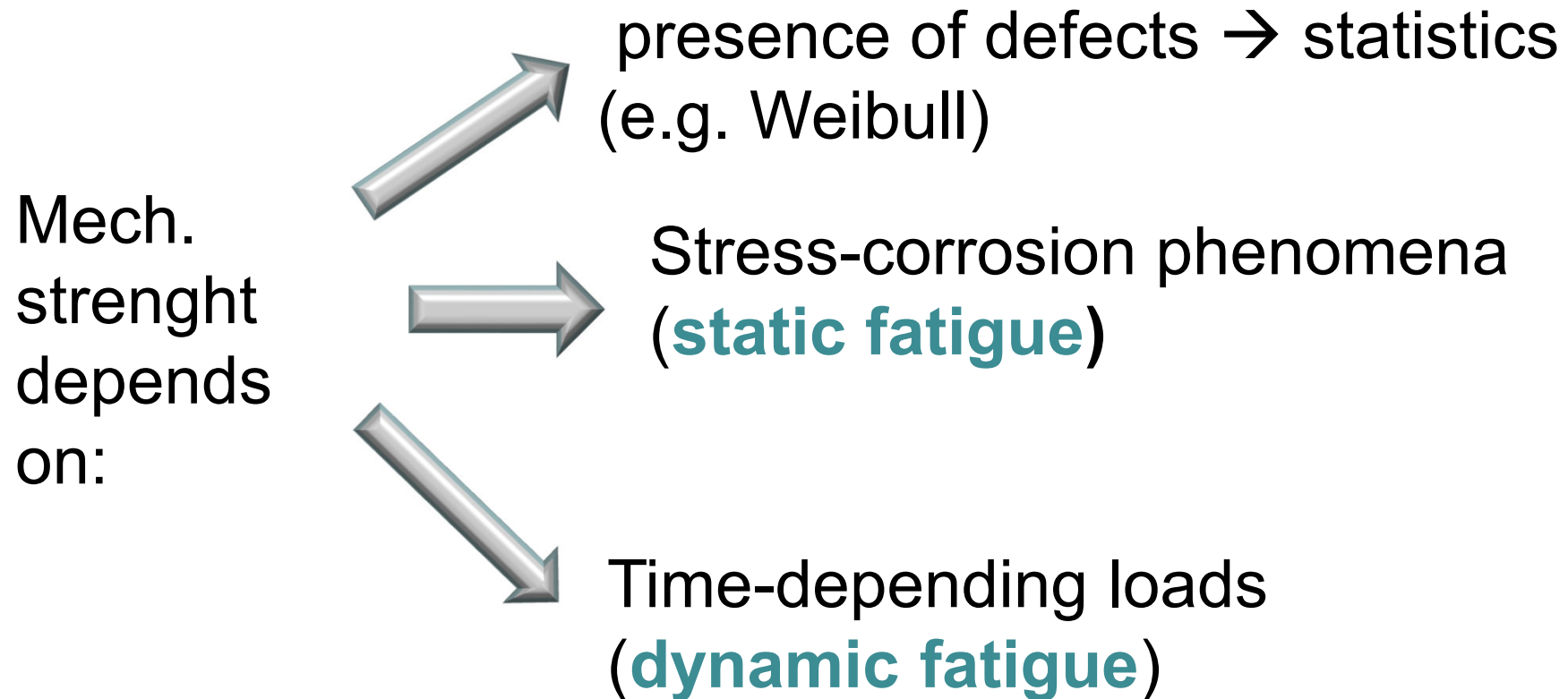
The mechanical strength of glasses and many ceramics tends to progressively decrease over time

- After a certain time, glass breaks due to a static stress which is lower than the expected one.
- Thus: the defect population evolves with time

This is a phenomenon known as **static fatigue**.

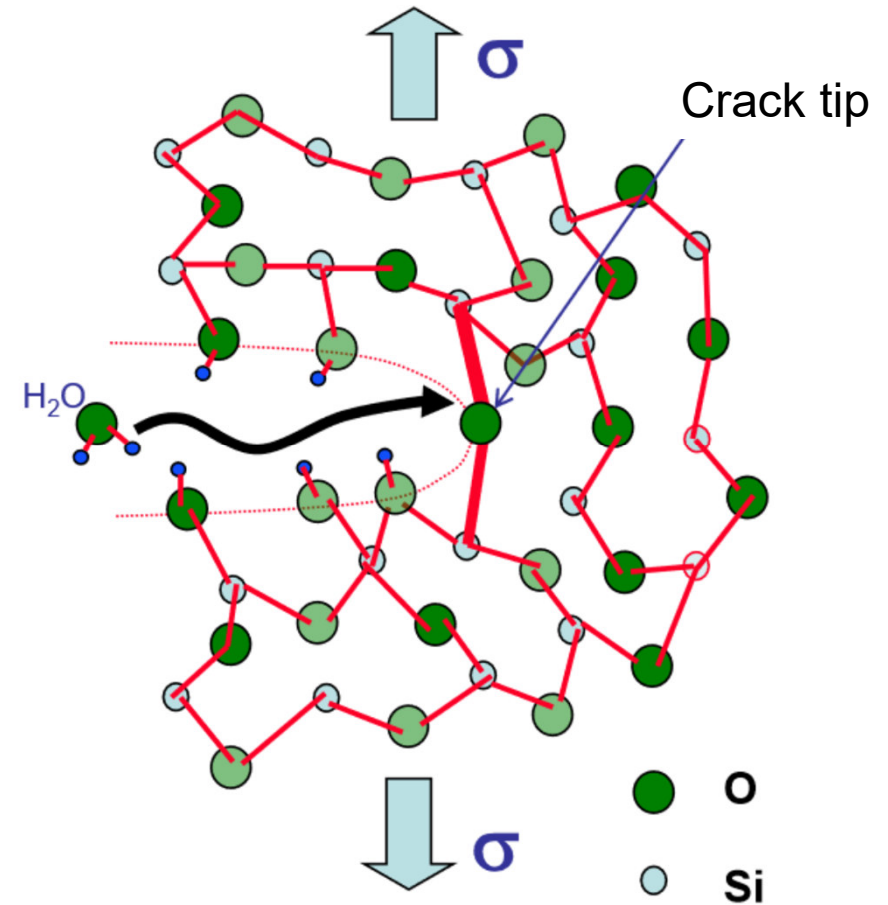


Effects of time need to be taken into account



In glasses and silicon oxide-based ceramics, water molecules break the deformed bonds at the tip of the crack.

- There is an energy barrier to overcome (threshold) in order to break the bonds
- Synergistic effects of stress and chemical attack on bonds
- Crack propagation is limited by the diffusion rate of the water (or aggressive species) at the tip of the crack.



If the bridging oxygen (see crack tip) is removed $\rightarrow$ crack extension

Moist environment $\rightarrow$ crack propagation even if $K < K_{IC}$

Where the bridging oxygen Si–O–Si is removed, the glass network is interrupted (as it is interrupted along the crack) → corrosion and removal of the bridging oxygen involves the extension of the crack (subcritical propagation of crack).

If a load is applied that opens the crack → accelerated propagation

